Modular forms I. Definitions and examples

1. Modular group Some notations: $\mathcal{H} = \{\tau \mid \mathrm{Im}\,\tau > 0\}$ $SL_2(\mathbb{Z}) = \{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a,b,c,d \in \mathbb{Z},\ ad-bc=1\}$

Def I 1.1. (1) Let $k \in \mathbb{Z}$ and $f: \mathcal{H} \to \mathbb{C}$ be meromorphic. f is weakly modular of weight k iff
$$f(\gamma(\tau)) = f(\tau)\,(c\tau+d)^k \quad \text{for} \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \text{ and } \tau \in \mathcal{H}$$

(2) $f: \mathcal{H} \to \mathbb{C}$ is a modular form of weight k iff
- ① f is holomorphic at $\mathcal{H}$
- ② f is holomorphic at ∞
- ③ f is weakly modular of weight k

The set of such functions is denoted by $M_k(SL_2(\mathbb{Z}))$

(3) f is a cusp form iff $f \in M_k(SL_2(\mathbb{Z}))$ and after expanding f to the Fourier series, i.e. $f(\tau) = \sum_{n \geq 1} a_n q^n$, $q = e^{2\pi i \tau}$, $a_0 = 0$.
The set of such functions is denoted by $S_k(SL_2(\mathbb{Z}))$

Notation ① To write it conveniently, we define weight-k operator $[\gamma]_k$ on f
by $(f[\gamma]_k)(\tau) = j(\gamma,\tau)^{-k} f(\gamma(\tau))$, $\gamma \in SL_2(\mathbb{Z})$, $\tau \in \mathcal{H}$
where $j(\gamma,\tau) := c\tau + d$, for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, $\tau \in \mathcal{H}$
Therefore, (1) is equivalent to $f[\gamma]_k = f \quad \forall\, \gamma \in SL_2(\mathbb{Z})$

From the definition of $j(\gamma,\tau)$ we immediately get the following properties:

Lemma I.1.2. (a) $j(\gamma\gamma',\tau) = j(\gamma,\gamma'(\tau))\,j(\gamma',\tau)$
(b) $(\gamma\gamma')(\tau) = \gamma(\gamma'(\tau))$
(c) $[\gamma\gamma']_k = [\gamma]_k[\gamma']_k$
(d) $\mathrm{Im}(\gamma(\tau)) = \dfrac{\mathrm{Im}(\tau)}{|j(\gamma,\tau)|^2}$
(e) $\dfrac{d\gamma(\tau)}{d\tau} = (j(\gamma,\tau))^{-2}$
Proof: (skip, it's just some computation) □.

From the above lemma and some computation, we have

Prop. I.1.3. (1) $M_k(SL_2(\mathbb{Z}))$ is a $\mathbb{C}$-vector space, $S_k(SL_2(\mathbb{Z}))$ is the subspace of $M_k(SL_2(\mathbb{Z}))$.
Moreover, $M(SL_2(\mathbb{Z})) = \bigoplus_k M_k(SL_2(\mathbb{Z}))$, $S(SL_2(\mathbb{Z})) = \bigoplus_k S_k(SL_2(\mathbb{Z}))$.

Later, we (probably) will prove the dimension formula: For even k, (odd is more complicated)
$$M_k(SL_2(\mathbb{Z})) = \begin{cases} \mathbb{C} & k = 0 \\ \{0\} & k \neq 0;\ k < 4;\ 2 \mid k \\ S_k(SL_2(\mathbb{Z})) \oplus \mathbb{C}E_k & \text{for even } k \geq 4. \end{cases}$$

$E_k(\tau) = \sum_{(c,d) \in \mathbb{Z}^2 - (0,0)} \frac{1}{(c\tau+d)^k}$, $\frac{1}{2\zeta(k)}$ $(k \geq 3)$ is the Eisenstein series.

$$\dim(S_k(SL_2(\mathbb{Z}))) = \begin{cases} \left\lfloor \frac{k}{12} \right\rfloor - 1 & \text{if } k \equiv 2 \pmod{12} \\ \left\lfloor \frac{k}{12} \right\rfloor & \text{otherwise} \end{cases}$$

(3) The ring $M(Sl_2(\mathbb{Z})) = \mathbb{C}(E_4, E_6)$, $S(Sl_2(\mathbb{Z})) = \Delta\, M(Sl_2(\mathbb{Z}))$

Δ is the discriminant function $\Delta(\tau) = (g_2(\tau))^3 - 27(g_3(\tau))^2$ where $g_2(\tau) = 60 G_4(\tau)$, $g_3(\tau) = 140 G_6(\tau)$

Some examples of modular forms (or not)

Eg I.1.3 $G_k(\tau) = \sum_{(c,d)\in\mathbb{Z}^2-(0,0)} \frac{1}{(c+d\tau)^k}$ $(k>4, 2|k)$, $\tau\in\mathcal{H}$ $\left(\sum'_{(c,d)} \frac{1}{(c+d\tau)^k}\right)$

$E_k(\tau) = \frac{1}{2\zeta(k)} G_k(\tau)$. are modular forms.

proof: ① Holomorphic: $\forall A, B>0$, define compact set $\Omega := \{\tau\in\mathcal{H} \mid |Re(\tau)|\le A,\ Im(\tau)\ge B\}$

(Sketch) We have: $\exists C>0$. s.t. $\forall \tau\in\Omega, \delta\in\mathbb{R}$, $|\tau+\delta| > C \sup\{1, |\delta|\}$ (*)

Indeed, (*) $\Leftrightarrow C < \left|\frac{\tau+\delta}{\delta}\right|$ $(|\delta|>1)$ and $C < |\tau+\delta|$ $(|\delta|<1)$

(It's easy to construct such C)

Then $|G_k(\tau)| < \sum'_{(c,d)} \frac{C}{(\sup(|c|,|d|))^k} < \infty$ on Ω

$\Rightarrow G_k(\tau)$ converges absolutely and uniformly on Ω $\Rightarrow G_k(\tau)$ is holomorphic on $\mathcal{H}$, Moreover, $G_k(\tau+n) = G_k(\tau) \Rightarrow |G_k(\tau)| < M$ (for some M)

$\Rightarrow G_k(\tau)$ is holomorphic at ∞

② $\forall \gamma\in Sl_2(\mathbb{Z}) := \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. we have $G_k(\gamma(\tau)) = (c\tau+d)^k \sum'_{(c',d')} \frac{1}{((c'a+d'c)\tau + (c'b+d'd))^k}$

$= (c\tau+d)^k \sum'_{(c',d')} \frac{1}{(c'\tau+d')^k}$

③ From the complex analysis, we know that

$\sum_{d\in\mathbb{Z}} \frac{1}{\tau+d} = \pi\cot\pi\tau = \pi i - 2\pi i \sum_{m\ge 0} q^m$ $q = e^{2\pi i\tau}$

Differentiating k times $\Rightarrow$ $\sum_{d\in\mathbb{Z}} \frac{1}{(\tau+d)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m\ge 1} m^{k-1} q^m$.

For even $k>2$, $\sum'_{(c,d)} \frac{1}{(c\tau+d)^k} = \sum_{d\ne 0} \frac{1}{d^k} + 2\sum_{c\ge 1} \sum_{d\in\mathbb{Z}} \frac{1}{(c\tau+d)^k}$

$= 2\zeta(k) + 2\cdot \frac{(2\pi i)^k}{(k-1)!} \sum_{c\ge 1} \sum_{m\ge 1} m^{k-1} q^{cm}$.

Therefore $G_k(\tau) = 2\zeta(k) + 2\cdot \frac{(2\pi i)^k}{(k-1)!} \sum_{n\ge 1} \sigma_{k-1}(n) q^n$ $(k>2, 2|k)$

$\sigma_k(n) = \sum_{\substack{m|n \\ m>0}} m^k$.

Eg.I.1.4. (An example of the cusp form). Let $g_2(\tau) = 60 G_4(\tau)$, $g_3(\tau) = 140 G_6(\tau)$, $\Delta(\tau) = (g_2(\tau))^3 - 27(g_3(\tau))^2$

Expand Δ to the fourier series, $a_0 = (60\cdot 2\zeta(4))^3 - 27(140\cdot 2\zeta(6))^2 = 0$, $a_1 \ne 0$

So Δ is a cusp form

g.I.1.5. The modular function $j: \mathcal{H} \to \mathbb{C}$ $\tau \mapsto 1728 \frac{(g_2(\tau))^3}{\Delta(\tau)}$

satisfies $(j[\gamma])(\tau) = j(\tau)$ $\forall \gamma\in Sl_2(\mathbb{Z})$, $\tau\in\mathcal{H}$

where after expanding j, we have $j(\tau) = \frac{(2\pi)^{12}+\dots}{(2\pi)^{12} q + \dots} = \frac{1}{q} + \dots$ which has a pole at 0

e.g. I.1.6. (4-square problem) Define the representation number

$r(n,k) := \#\{v \in \mathbb{Z}^k \mid n = v_1^2 + \dots + v_k^2\}$. and the function $\theta(\tau,k) = \sum_{n\geq 0} r(n,k) q^n$. $q = e^{2\pi i \tau}$

θ has the following properties: ① $\theta(\tau, k_1+k_2) = \theta(\tau,k_1)\theta(\tau,k_2)$ Let $\theta(\tau)_1 = \theta(\tau,1) = \sum_{d\in\mathbb{Z}} e^{2\pi d^2 \tau}$

② $\theta(\tau+1,k) = \theta(\tau,k)$

③ Using Poisson summation formula: $\theta\left(\frac{-1}{4\tau}\right) = \theta(\tau)\cdot\sqrt{-2i\tau}$.

From this, we have $\theta\left(\frac{\tau}{4\tau+1}, 4\right) \overset{\theta}{=} \left(\sqrt{4\tau+1}\ \theta(\tau)\right)^4 \overset{①}{=} (4\tau+1)^2 \theta(\tau,4)$

That means $\theta[\gamma]_2(\tau) = \theta(\tau,4)$ for $\gamma = \pm\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ & $\pm\begin{bmatrix}1 & 0\\ 4 & 1\end{bmatrix}$ which generalizes

$\Gamma(4) = \left\{\begin{pmatrix}a & b\\ c & d\end{pmatrix} \in SL_2(\mathbb{Z}) \,\middle|\, \begin{pmatrix}a & b\\ c & d\end{pmatrix} \equiv \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \bmod 4\right\}$ ∎

Def I.1.7. Let $N>0$ ① The principal congruence group of level N $\Gamma(N) := \left\{\begin{pmatrix}a & b\\ c & d\end{pmatrix} \in SL_2(\mathbb{Z}) \,\middle|\, \dots \equiv \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \bmod\right.$

② The congruence group Γ is level N iff $\Gamma(N) \subseteq \Gamma$.

③ $\Gamma_0(N) = \left\{A \in SL_2(\mathbb{Z}) \,\middle|\, A \equiv \begin{pmatrix}* & *\\ 0 & *\end{pmatrix} \bmod N\right\}$, $\Gamma_1(N) = \left\{A \in SL_2(\mathbb{Z}) \,\middle|\, A \equiv \begin{pmatrix}1 & *\\ 0 & 1\end{pmatrix} \bmod N\right\}$.

$\Rightarrow \Gamma(N) \trianglelefteq \Gamma_1(N) \trianglelefteq \Gamma_0(N) \leq SL_2(\mathbb{Z})$.

Thm I.1.8. ① $\Gamma(N) \trianglelefteq SL_2(\mathbb{Z})$. and $SL_2(\mathbb{Z})/\Gamma(N) \cong SL_2(\mathbb{Z}/N\mathbb{Z})$, $[SL_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p\mid N}\left(1-\frac{1}{p^2}\right)$.

② $\Gamma(N) \trianglelefteq \Gamma_1(N)$ and $\Gamma_1(N)/\Gamma(N) \cong \mathbb{Z}/N\mathbb{Z}$

③ $\Gamma_1(N) \trianglelefteq \Gamma_0(N)$ and $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^*$

proof: (sketch) ① Define $\pi : SL_2(\mathbb{Z}) \longrightarrow SL_2(\mathbb{Z}/N\mathbb{Z})$

② Define $\pi : \Gamma_1(N) \longmapsto \mathbb{Z}/N\mathbb{Z}$ $\begin{pmatrix}a & b\\ c & d\end{pmatrix} \mapsto b \bmod N$

③ Define $\pi : \Gamma_0(N) \longrightarrow (\mathbb{Z}/N\mathbb{Z})^*$ $\begin{pmatrix}a & b\\ c & d\end{pmatrix} \mapsto d \bmod N$ ∎.

Def I.1.9. Γ be the congruence group of $SL_2(\mathbb{Z})$ and $k \in \mathbb{Z}$. $f : \mathcal{H} \to \mathbb{C}$ is a modular form of weight k with respect to Γ if

(1) $f \in H(\mathcal{H})$; (2) $f[\gamma]_k = f$ $\forall \gamma \in \Gamma$; (3) $f[\alpha]_k$ is holomorphic at ∞, $\forall \alpha \in SL_2(\mathbb{Z})$

Moreover, if $a_0 = 0$, then f is a cusp form ... . Similarly we have notations: $M_k(\Gamma)$, $S_k(\Gamma)$.